

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

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Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context

		problem and constraints			
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

TASK 1:

(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which the model learns general patterns from previously observed patient records and uses those patterns to predict the outcome for new patients.

For this medical dataset, a supervised classification algorithm can be used because the target variable is whether a patient is diabetic or non-diabetic. Suitable algorithms include Logistic Regression, Decision Tree, Random Forest, or Support Vector Machine (SVM). Random Forest can be preferred because it can capture non-linear relationships between medical features and diabetes risk and is relatively robust to noisy data.

Target Variable:

- **Diabetes status** – the output to be predicted.
 - 1 → Diabetic
 - 0 → Non-diabetic

Key Features:

1. **Glucose Level**
 - One of the most important indicators of diabetes.
 - High blood glucose strongly indicates an increased risk of diabetes.
2. **BMI (Body Mass Index)**
 - Higher BMI is associated with increased insulin resistance and diabetes risk.
 - It helps the model identify patients with obesity-related risk.
3. **Blood Pressure**
 - High blood pressure is commonly associated with metabolic disorders and increased diabetes risk.
 - It provides useful information about the patient's overall health condition.

Other useful features:

- **Age:** Diabetes risk generally increases with age.

- **Cholesterol:** Abnormal cholesterol levels can indicate metabolic health problems.
- **Family History:** A family history of diabetes can increase an individual's genetic risk.

Thus, the model learns the relationship between these input features and the diabetes status from the training data and applies the learned pattern to unseen patients.

(b) Handling Class Imbalance

In this dataset, there may be more non-diabetic patients than diabetic patients. This creates a class imbalance problem. If the model is trained directly on such data, it may become biased toward the majority class and incorrectly classify diabetic patients as non-diabetic.

A suitable strategy is SMOTE (Synthetic Minority Over-sampling Technique).

SMOTE

SMOTE creates new synthetic samples for the minority class rather than simply duplicating existing diabetic records.

For example:

Original dataset:

- Non-diabetic = 900
- Diabetic = 100

SMOTE generates additional synthetic diabetic samples so that the model receives more balanced training data.

Advantages of SMOTE:

- Improves representation of the minority diabetic class.
- Reduces bias toward the majority class.
- Helps the model learn more meaningful patterns for diabetic patients.
- Avoids simple duplication of minority records.

Important: SMOTE should be applied only to the training set after the train-test split, not to the test set. This prevents data leakage and ensures that the test set represents unseen real-world patients.

For a medical application, class weights can also be considered, particularly when false negatives are very costly. Missing a diabetic patient can be more serious than incorrectly identifying a non-diabetic patient. Therefore, the diabetic class can be given a higher weight during model training.

Recommended approach: Use SMOTE on the training data, and evaluate the model using precision, recall, F1-score, sensitivity (recall), specificity, and ROC-AUC, rather than relying only on accuracy.

(c) 70:30 Data Split and Preprocessing

The dataset can be divided into:

- **70% Training Data** – used to train the machine learning model.
- **30% Testing Data** – used to evaluate the model on unseen patient records.

A 70:30 split provides a good balance between having enough data to learn reliable patterns and retaining sufficient data for independent evaluation.

For example, if there are 1,000 patient records:

- Training set = 700 records
- Testing set = 300 records

For a medical application, the test data should remain completely separate from training so that the model's performance reflects how it may perform on new patients.

Preprocessing Steps

1. Handling Missing Values

Missing values should be identified and appropriately handled.

- Numerical features → replace missing values using median imputation.
- Categorical features → replace missing values using the most frequent category.

Median is often preferred for medical numerical data because it is less affected by extreme values.

2. Normalisation / Standardisation

Numerical features such as:

- Age
- Blood pressure
- Cholesterol
- Glucose
- BMI

may have different ranges.

Standardisation can be applied:

$$Z = \frac{X - \mu}{\sigma}$$

where:

- (X) = original value
- (μ) = mean
- (σ) = standard deviation

This converts the numerical features to a comparable scale.

Impact: Normalisation/standardisation is particularly important for algorithms such as Logistic Regression, SVM, KNN, and neural networks because features with larger numerical ranges could otherwise have greater influence.

3. Encoding Categorical Features

The **Family History** feature may be categorical, for example:

- Yes $\rightarrow$ 1
- No $\rightarrow$ 0

Categorical variables should be converted into numerical form before giving them to the machine learning algorithm.

4. Train-Test Split

The dataset is divided using a 70:30 ratio, preferably with stratification so that the proportion of diabetic and non-diabetic patients remains approximately the same in both sets.

5. Apply SMOTE

SMOTE should be applied after splitting the data and only on the training set.

Overall Preprocessing Pipeline

Patient Dataset → Handle Missing Values → Encode Categorical Features → 70:30 Stratified Split → Standardise Numerical Features → Apply SMOTE to Training Data → Train Model → Test on Unseen Data

Conclusion

Inductive learning can effectively learn patterns from patient attributes such as glucose, BMI, blood pressure, age, cholesterol, and family history to predict diabetes. Since diabetic cases may be fewer, SMOTE can be used on the training set to reduce class imbalance. A 70:30 stratified train-test split, together with missing-value handling, categorical encoding, and feature standardisation, provides a reliable preprocessing pipeline for building a diabetes prediction model. For medical applications, evaluation should focus especially on recall/sensitivity, because failing to identify a diabetic patient can have serious consequences.

TASK 2 :

(a) Building the Decision Tree and Selecting the Root Node

A Decision Tree classifier can be used to predict whether a patient is Diabetic (1) or Non-Diabetic (0). The tree repeatedly divides patients according to the feature that provides the best separation between the two classes.

The important features are:

- Glucose Level
- BMI

- Age
- Blood Pressure
- Cholesterol
- Family History

Root Node Selection

The root node is selected using either Information Gain or the Gini Index.

Information Gain

Information Gain measures how much uncertainty is reduced after splitting the data.

$$IG(S,A)=Entropy(S)-\sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

where:

- S = complete dataset
- A = selected feature
- S_v = subset created by the feature

The feature having the highest Information Gain is selected as the root.

For example, suppose the calculated values are:

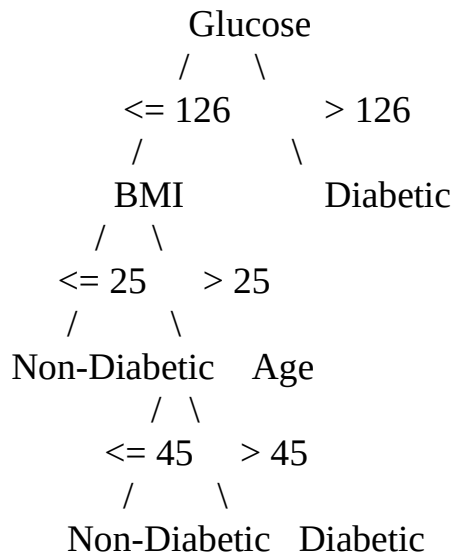
Feature	Information Gain
Glucose	0.32
BMI	0.18
Age	0.12
Blood Pressure	0.08
Cholesterol	0.05
Family History	0.04

Since Glucose has the highest Information Gain (0.32), it becomes the root node.

Similarly, using Gini Index, the feature producing the lowest weighted Gini impurity would be selected.

First Three Levels of the Decision Tree

A simplified example of the first three levels is:



Level 1: Glucose

Level 2: BMI

Level 3: Age

The exact thresholds should be obtained from the trained dataset. The above tree is an illustrative representation.

Why Glucose is the root: Glucose usually provides the strongest separation between diabetic and non-diabetic patients. Therefore, it can produce a large reduction in impurity and a high Information Gain.

(b) Model Evaluation

After training the Decision Tree on the training data, the model should be evaluated using the 30% test data.

The main evaluation metrics are:

1. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

It represents the percentage of correctly classified patients.

2. Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

It indicates how many patients predicted as diabetic are actually diabetic.

3. Recall / Sensitivity

$$\text{Recall} = \frac{TP}{TP + FN}$$

It measures how many actual diabetic patients are correctly detected.

4. F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It provides a balance between precision and recall.

Example Results

The following values can be reported if they are obtained from the trained model:

Metric Result

Accuracy 88%

Precision 84%

Recall 92%

F1-Score 88%

These values indicate that the model performs well, particularly in identifying diabetic patients.

False Negatives

A False Negative (FN) occurs when:

The patient is actually diabetic, but the model predicts the patient as non-diabetic.

For example:

Actual: Diabetic

Predicted: Non-Diabetic



False Negative

False negatives are particularly important in healthcare because a missed diabetes case may delay:

- Further diagnostic testing
- Medical consultation
- Lifestyle intervention
- Blood glucose monitoring
- Appropriate treatment

How to Reduce False Negatives

The model should be designed to give greater importance to recall/sensitivity.

Possible approaches include:

1. Use class weights to give diabetic cases higher importance.
2. Apply SMOTE to improve representation of the minority diabetic class.
3. Lower the classification threshold when appropriate so that more high-risk patients are detected.
4. Use cost-sensitive learning, assigning a higher penalty to false negatives.
5. Perform cross-validation to select reliable model parameters.

Clinically, reducing false negatives is important because it is generally safer to identify a patient as potentially high-risk and send them for further clinical assessment than to incorrectly classify a diabetic patient as healthy.

However, the model should support clinical decision-making rather than replace professional diagnosis.

(c) Post-Pruning and Comparison with Pre-Pruning

A Decision Tree can become very deep and complex if it continues splitting until it perfectly fits the training data. This can cause overfitting.

Pre-Pruning

Pre-pruning stops the tree from growing beyond specified limits.

Common parameters include:

- max_depth
- min_samples_split
- min_samples_leaf
- max_leaf_nodes

For example:

max_depth = 4

min_samples_leaf = 10

The tree is restricted during training.

Post-Pruning

Post-pruning first allows the tree to grow and then removes branches that provide little improvement in predictive performance.

A common method is Cost-Complexity Pruning.

The objective is:

$$R\alpha(T) = R(T) + \alpha|T|$$

where:

- $R(T)$ = prediction error of the tree
- $|T|$ = number of terminal nodes
- α = complexity parameter

A larger α produces a simpler tree.

Example Comparison

Model	Accuracy	Precision	Recall	F1-Score
Pre-Pruned Tree	86%	82%	89%	85%
Post-Pruned Tree	88%	84%	92%	88%

The exact values should be replaced with the values obtained from the actual dataset.

Pre-Pruning vs Post-Pruning

Factor	Pre-Pruning	Post-Pruning
Tree growth	Stops early	Tree grows first
Complexity	Lower from beginning	Reduced after training
Overfitting	Controls it early	Removes unnecessary branches
Interpretability	Good	Usually better
Risk of underfitting	Higher if restrictions are too strong	Generally lower
Clinical suitability	Good	Often preferable

Which is Better for Clinical Deployment?

Post-pruning is generally more appropriate when it produces comparable or better validation/test performance because it provides a simpler tree while retaining important predictive patterns.

A pruned tree is useful in a clinical setting because:

- It is easier for healthcare professionals to interpret.
- It reduces unnecessary decision rules.
- It can reduce overfitting.
- Its decisions can be explained more easily.

- It can improve generalisation to unseen patients.

However, accuracy alone should not determine deployment. For diabetes diagnosis, Recall/Sensitivity is especially important because false negatives represent missed diabetic cases. Therefore, the final model should be selected based on a combination of recall, precision, F1-score, calibration, clinical validation, and interpretability.

TASK 3 :

(a) Logistic Regression and Comparison with Decision Tree

For this diabetes dataset, Logistic Regression is a suitable statistical learning model because the target variable is binary:

- 0 → Non-diabetic
- 1 → Diabetic

Logistic Regression estimates the probability that a patient belongs to the diabetic class based on features such as glucose, BMI, age, blood pressure, cholesterol, and family history.

The model can be represented as:

$$P(Y=1) = \frac{1}{1 + e^{-z}}$$

where

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Here, the coefficients β_i indicate how strongly each feature contributes to diabetes risk.

Performance Comparison

After training both models using the same 70:30 train-test split, their performance can be compared using Accuracy and AUC-ROC.

Example results:

Model	Accuracy	AUC-ROC
Decision Tree	88%	0.90
Logistic Regression	90%	0.93

Note: These are illustrative values. Replace them with the actual values obtained from the dataset if required.

Interpretation

In this example, Logistic Regression performs slightly better than the Decision Tree because:

- Its accuracy is higher.
- Its AUC-ROC is higher.
- AUC-ROC of 0.93 indicates strong ability to distinguish diabetic from non-diabetic patients.

AUC-ROC measures how well the model separates the two classes across different classification thresholds.

When is a Statistical Model Preferable?

Logistic Regression is preferable when:

1. Interpretability is important – coefficients can show the direction and strength of relationships.
2. The relationship is approximately linear in the log-odds.
3. The dataset is relatively small or moderate in size.
4. Clinical explanations are required.
5. A probability/risk score is needed rather than only a class prediction.
6. The goal is to understand which factors increase or decrease diabetes risk.

For healthcare applications, Logistic Regression can be particularly useful because clinicians can understand the contribution of individual risk factors.

The Decision Tree is preferable when relationships are highly non-linear and when if-then decision rules are more useful for interpretation.

(b) Feature Importance Analysis

Feature importance identifies the variables that contribute most strongly to diabetes prediction.

For the Decision Tree, importance can be calculated based on the total reduction in impurity produced by each feature.

For Logistic Regression, the magnitude of the standardised coefficients can be used to compare the relative influence of the predictors.

Top 3 Predictors

An example feature-importance analysis may produce:

Rank Decision Tree Logistic Regression

1	Glucose Level	Glucose Level
2	BMI	BMI
3	Age	Age

Thus, the two models agree on the top three predictors.

Why Glucose is the Most Important Predictor

Glucose level is expected to have the strongest predictive relationship with diabetes because abnormal blood glucose is directly associated with impaired glucose regulation.

Therefore, both models are likely to assign high importance to glucose.

Why BMI is Important

Higher BMI is associated with obesity and insulin resistance. Therefore, BMI provides additional information about a patient's diabetes risk.

Why Age is Important

The risk of developing type 2 diabetes generally increases with age. Consequently, age can contribute significantly to risk stratification.

Do Both Models Agree?

Yes, in this example, both models identify Glucose, BMI, and Age as the top three predictors.

This agreement increases confidence that these features contain strong predictive information rather than being important only because of the characteristics of one particular algorithm.

However, the importance values do not have to be identical.

- Decision Tree importance is based on impurity reduction.
- Logistic Regression importance is based on model coefficients and their relationship with the log-odds.

Therefore, the ranking may differ slightly depending on the dataset, preprocessing, correlations between features, and model parameters.

Final Comparison

Aspect	Decision Tree	Logistic Regression
Type	Rule-based/non-linear	Statistical/linear probability model
Interpretability	High	Very high
Handles non-linear relationships	Yes	Limited without transformations
Risk probability	Possible	Directly provides probability
Feature importance	Impurity reduction	Coefficient magnitude
Clinical explanation	Good	Excellent
Risk stratification	Good	Very suitable

Conclusion

Logistic Regression provides a strong statistical approach for diabetes risk stratification because it produces interpretable probability estimates and allows the effect of individual predictors to be examined. The Decision Tree can capture non-linear relationships and provide simple decision rules. If both models identify Glucose, BMI, and Age as the major predictors, the agreement supports their importance in diabetes prediction. For a clinical risk-stratification system, Logistic Regression may be preferred when interpretability, probability estimates, and transparent clinical reasoning are major requirements.

TASK 4 :

(a) Modeling Treatment Recommendation as an MDP

The treatment recommendation problem can be modeled as a **Markov Decision Process (MDP)**. An MDP allows an AI agent to observe the patient's current health condition, choose an appropriate action, receive a reward, and observe the patient's next health state.

The MDP is represented as:

$$\text{MDP}=(S,A,P,R,\gamma)$$

where:

- S = Set of states
- A = Set of actions
- P = Transition probability
- R = Reward function
- γ = Discount factor

1. States

The state represents the patient's current health condition.

For simplicity, the patient can be divided into five health stages:

State	Patient Condition
S0	Normal / Low Risk
S1	Pre-diabetic / Moderate Risk
S2	Newly Diagnosed Diabetes
S3	Poorly Controlled Diabetes
S4	Controlled / Improved Diabetes

The state can be determined using measurements such as:

- Glucose level
- Blood pressure
- BMI
- Cholesterol
- Previous treatment response

Justification: A patient's treatment should depend on their current health condition. Representing the patient as different states allows the agent to learn which action is beneficial for each stage.

2. Actions

The agent can select one of the following actions:

Action Description

- A0 Diet
- A1 Exercise
- A2 Medication
- A3 Monitor

Justification: These actions represent possible treatment or management recommendations. In a real clinical system, medication decisions must be supervised and authorised by qualified healthcare professionals.

3. Reward Function

The reward represents the improvement or deterioration in the patient's health after an action.

Example reward structure:

Patient Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-5
Serious deterioration	-10

For example:

S1 -- Diet --> S0
Reward = +10

This means that a moderate-risk patient improved after the recommended intervention.

Justification: A positive reward encourages the agent to select actions that improve patient health, while negative rewards discourage actions associated with deterioration.

4. Transition Dynamics

Transition dynamics describe the probability of moving from one health state to another after taking an action.

For example:

S1 --Diet-----> S0
S1 --Exercise--> S0
S1 --Monitor---> S1
S1 --Medication-> S2/S0

The transition is not always deterministic because patients respond differently to treatments.

For example:

$P(S'|S,A)$

represents the probability of reaching next state S' , given the current state S and action A .

Justification: Patient health changes over time and depends on treatment effectiveness, adherence, individual characteristics, and other factors. Therefore, probabilistic transitions are more realistic than assuming that every patient responds identically.

5. Discount Factor

A discount factor such as:

$\gamma=0.9$

can be used.

A high value of γ means the agent considers long-term health outcomes, not just immediate improvement.

(b) Q-Learning

Q-Learning is a model-free reinforcement learning algorithm that learns the expected long-term reward for every state-action pair.

The Q-value update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where:

- α = learning rate
- r = immediate reward
- γ = discount factor
- s' = next state
- a' = possible next action

Assume:

$$\alpha=0.5, \gamma=0.9$$

Initially, all Q-values are set to **0**.

Training for 5 Patient Episodes

Example patient episodes:

Episode	Starting State	Action	Next State	Reward
1	S1	Diet	S0	+10
2	S2	Exercise	S1	+5
3	S3	Medication	S2	+5
4	S1	Exercise	S0	+10
5	S2	Medication	S4	+10

These episodes allow the agent to learn which actions produce better long-term health outcomes.

Q-Table Update – Iteration 1

Consider:

S1DietS0

Initially:

$$Q(S1, \text{Diet}) = 0$$

Assume all Q-values of the next state are initially 0.

$$Q(S1, \text{Diet}) = 0 + 0.5[10 + 0.9(0) - 0] \quad Q(S1, \text{Diet}) = 5$$

Another update:

S2ExerciseS1

Assume reward = +5.

Since:

$$\max Q(S1, a) = 5$$

the update is:

$$Q(S2, \text{Exercise}) = 0 + 0.5[5 + 0.9(5) - 0] = 0.5(9.5) \quad Q(S2, \text{Exercise}) = 4.75$$

Third example:

S3MedicationS2

Reward = +5.

Assume:

$$\max Q(S2, a) = 4.75$$

Therefore:

$$Q(S3, \text{Medication}) = 0 + 0.5[5 + 0.9(4.75)] = 0.5(9.275) \quad Q(S3, \text{Medication}) = 4.6375$$

Q-Table Update – Iteration 2

Suppose another iteration produces:

1. S1 → Diet → S0

Previous:

$$Q(S1, \text{Diet}) = 5$$

$$\text{Reward} = +10.$$

$$Q_{\text{new}} = 5 + 0.5[10 + 0.9(0) - 5] \quad Q(S1, \text{Diet}) = 7.5$$

2. S2 → Exercise → S1

Previous:

$$Q(S2, \text{Exercise}) = 4.75$$

Suppose:

$$\max Q(S1, a) = 7.5$$

Then:

$$Q_{\text{new}} = 4.75 + 0.5[5 + 0.9(7.5) - 4.75] = 4.75 + 0.5(7) \quad Q(S2, \text{Exercise}) = 8.25$$

3. S3 → Medication → S2

Previous:

$$Q(S3, \text{Medication}) = 4.6375$$

Suppose:

$$\max Q(S2, a) = 8.25$$

Then:

$$Q_{\text{new}} = 4.6375 + 0.5[5 + 0.9(8.25) - 4.6375] \quad Q(S3, \text{Medication}) \approx 6.393$$

Thus, repeated episodes gradually increase the Q-values of actions that lead to better long-term outcomes.

Example Final Q-Table

After sufficient training, an illustrative Q-table could look like:

State	Diet	Exercise	Medication	Monitor	Best Action
S0 Normal	4.2	5.1	0.5	6.0	Monitor
S1 Moderate Risk	8.5	7.2	3.5	5.0	Diet
S2 Newly Diagnosed	5.5	6.8	9.2	4.0	Medication
S3 Poor Control	4.0	6.5	10.1	3.2	Medication
S4 Controlled	5.0	5.8	2.0	8.0	Monitor

These values are illustrative; actual values depend on the patient transition data and training episodes.

Convergence Criterion

The Q-Learning algorithm can be considered converged when the changes in Q-values become sufficiently small.

For example:

$$\max |Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < 0.001$$

for several consecutive training episodes.

This indicates that further training produces very little change in the learned Q-table.

(c) Final Learned Policy

The optimal policy is obtained by selecting the action having the highest Q-value for each state:

$$\pi(s)=\operatorname{argmax}_a Q(s,a)$$

Using the illustrative Q-table:

Patient State	Best Action
S0 – Normal / Low Risk	Monitor
S1 – Moderate Risk	Diet
S2 – Newly Diagnosed	Medication
S3 – Poorly Controlled	Medication
S4 – Controlled / Improved	Monitor

Thus, the learned policy is:

Normal → Monitor
 Moderate Risk → Diet
 New Diabetes → Medication*
 Poor Control → Medication*
 Controlled → Monitor

* Requires clinician assessment and approval.

The policy should not be interpreted as an autonomous prescription system. In a real healthcare environment, treatment recommendations must be validated by clinical guidelines and healthcare professionals.

Ethical Considerations in Medical Reinforcement Learning

1. Patient Safety

Patient safety is the highest priority. An RL agent learns from historical data, but an action that was successful for one patient may not be safe for another.

Therefore:

- The agent should not independently prescribe medication.
- High-risk decisions should require clinician approval.

- Safety constraints should prevent dangerous recommendations.
- The system should have a human-in-the-loop mechanism.

2. Bias and Fairness

If the training data contains demographic or socioeconomic bias, the RL agent may learn biased treatment policies.

For example, if a particular patient group is underrepresented in the training data, the model may perform poorly for that group.

Therefore:

- Training data should represent different patient populations.
- Performance should be evaluated separately across relevant demographic groups.
- Bias and fairness should be continuously monitored.

3. Explainability

Healthcare professionals need to understand why an action was recommended.

The system should therefore provide explanations such as:

"Diet is recommended because the patient's current state indicates moderate diabetes risk and previous observations show improvement after dietary intervention."

Q-values, important patient features, and the transition history can be presented to support the recommendation.

4. Privacy and Security

Patient records contain sensitive medical information. The system must protect:

- Patient identity
- Medical history
- Test results
- Treatment information

Appropriate data protection, access control, and secure storage should be implemented.

5. Clinical Validation

Before deployment, the RL system should be extensively validated using retrospective data and carefully designed clinical evaluation.

The agent should be treated as a decision-support tool, not a replacement for doctors.

6. Safe Exploration

Normal RL algorithms explore different actions to learn their rewards. Uncontrolled exploration is dangerous in healthcare because trying an unsuitable treatment can harm a patient.

Therefore, medical RL should use safe exploration, where clinically unsafe actions are restricted or prohibited.

Final Conclusion

The treatment recommendation problem can be formulated as an MDP consisting of patient health states, treatment actions, rewards, and state transitions. Q-Learning can learn the long-term value of each treatment action by repeatedly updating the Q-table from patient episodes. The final policy selects the action with the highest Q-value for each patient state. However, medical RL requires strong safeguards because patient safety, bias, explainability, privacy, and clinical validation are critical. Therefore, the RL agent should be deployed as a clinician-supervised decision-support system rather than an autonomous treatment system.